

☆ Amplitude Modulation

"Amplitude modulation is defined as the process in which the amplitude of the carrier signal is varied in accordance with the modulating signal or message signal."

→ Amplitude modulation may be defined as a system in which the maximum amplitude of the carrier wave is made proportional to the instantaneous value (amplitude) of the modulating or baseband signal.

high freq & fixed freq

modulating sig contain information to be transmitted

→ Consider a sinusoidal carrier signal $c(t)$ is defined as,

$$c(t) = A \cos \omega_c t \quad \text{--- (1)} \Rightarrow c(t) = A_c \cos(2\pi f_c t + \phi) \pm$$

where A = maximum amplitude of the carrier wave

ω_c = carrier frequency

ϕ = Phase angle.

For our convenience, assume the phase angle of the carrier signal is zero.

Let $x(t)$ = modulating signal or baseband signal

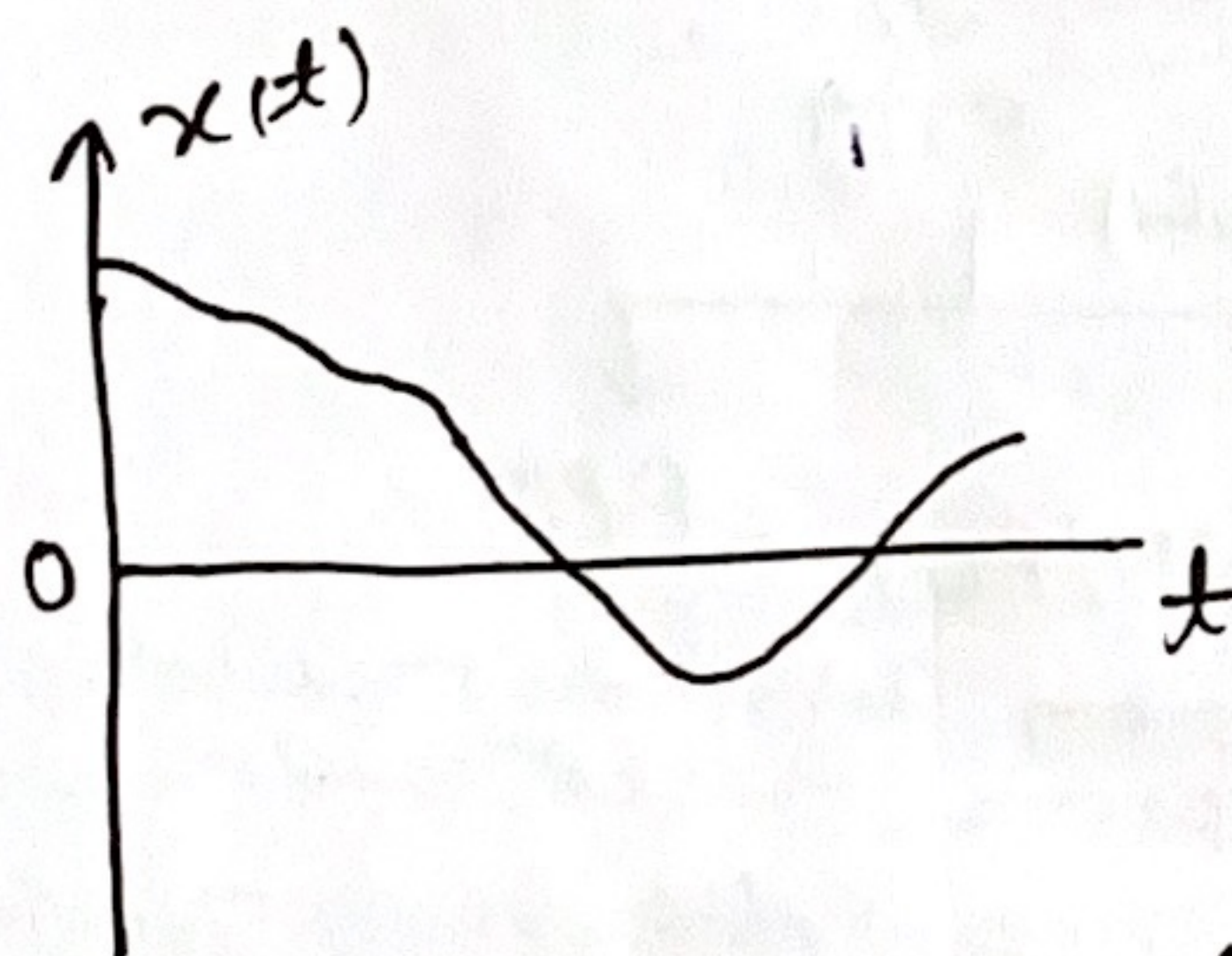
Then according to amplitude modulation, the max. amplitude A of the carrier wave will have to be made proportional to the instantaneous amplitude of modulating signal $x(t)$.

→ The standard equation for amplitude modulation (AM) wave may be expressed as,

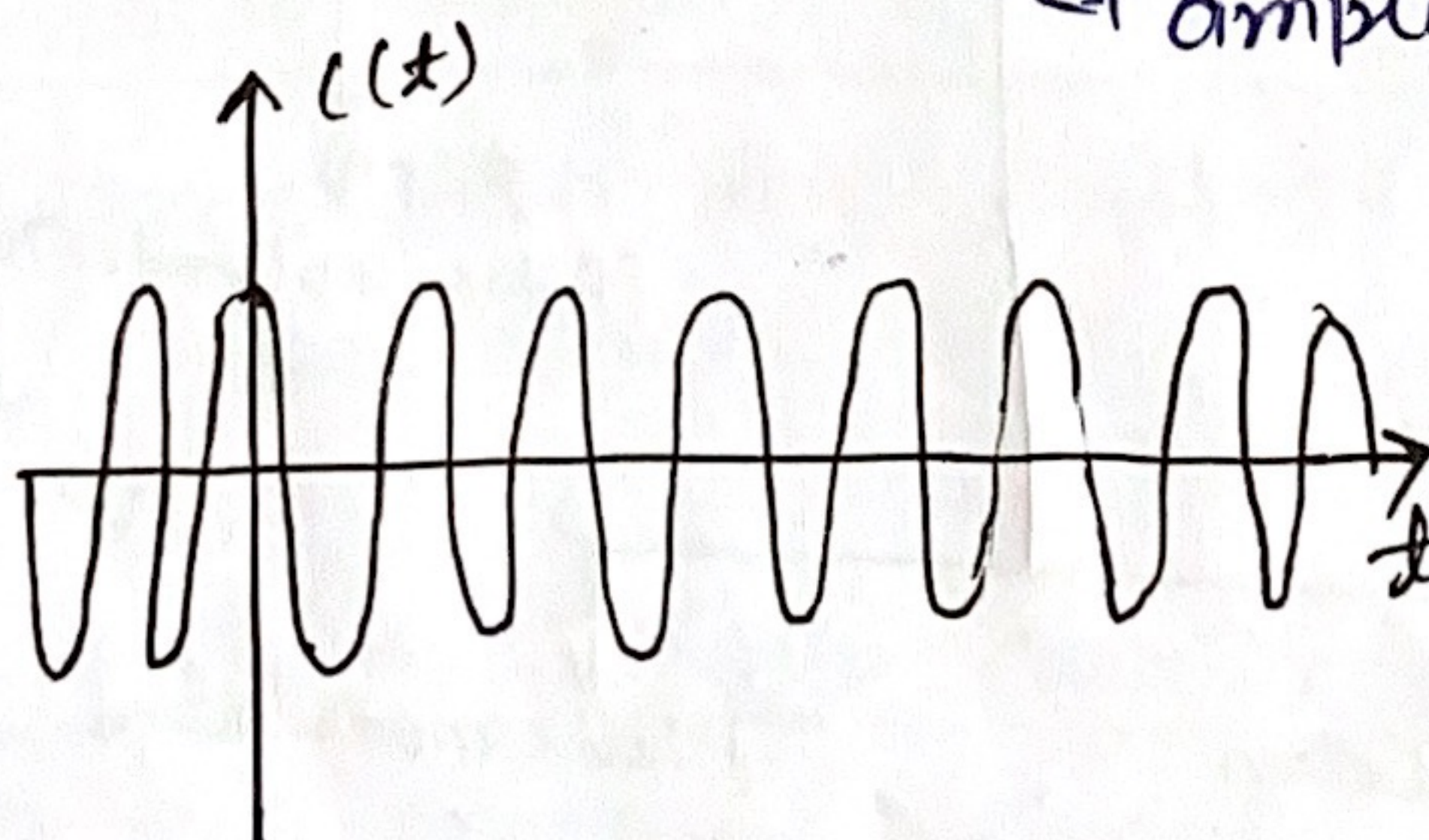
$$s(t) = x(t) \cos \omega_c t + A \cos \omega_c t \quad \text{--- (2)}$$

$$s(t) = [A + x(t)] \cos \omega_c t \quad \text{--- (3)}$$

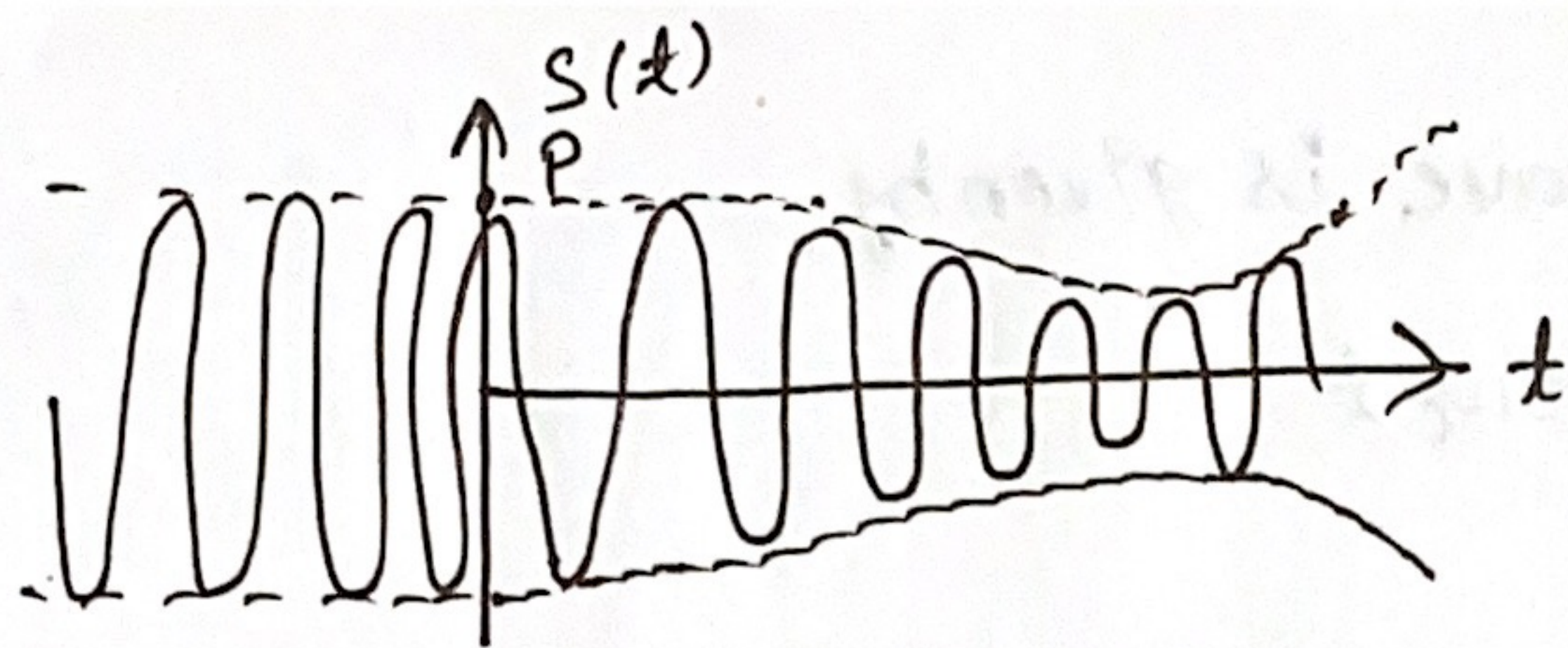
⇒ amplitude modulated wave.



(a) modulating signal



(b) carrier signal



modulating signal
or
Baseband signal
or
message signal

(c) modulated signal

e.g. ② and ③ shows the time-domain behaviour of amplitude-modulated signal.

- carrier signal (i.e. $c(t) = A \cos \omega_c t$) is a fixed frequency signal having frequency ω_c
- modulating signal, $x(t)$ contains the information to be transmitted.
- In amplitude modulation process, information is superimposed upon the carrier signal in the form of amplitude variations of the carrier signal.
- The resulting signal from the process of amplitude modulation is called amplitude modulated signal or AM wave.
- In amplitude modulation process, the frequency and phase of the carrier wave remains constant, whereas the max. amplitude of the carrier signal varies according to the instantaneous value of the information signal.
- amplitude modulated wave e.g. ③ has constant frequency ω_c and amplitude $A + x(t)$
- In Fig (c) the AM wave has a time-varying amplitude called as the envelope of the AM wave.
- Envelope of AM wave consists of modulating signal. or we can say that it has the same shape as the modulating sig.
- Fig (c) shows the process of amplitude modulation. We can see that upto point P, modulating signal is not applied so there is no modulation and max. amplitude of the carrier remains constant at A. Now, at point P, the modulating signal is applied. Therefore, after point P, the amplitude modulation occurs.

General expression of AM wave is given by,

$$S(t) = [A + x(t)] \cos \omega_c t$$

$$S(t) = E(t) \cos \omega_c t$$

where $E(t) = A + x(t)$

$E(t)$ = envelope of AM wave. and it consists of modulating s/g, $x(t)$

Hence, the modulating s/g may be recovered from an AM wave by detecting the envelop.

Spectrum of AM wave or Frequency-Domain Representation

→ Description of a signal w.r.t its frequency is called a frequency-domain representation.

→ Spectrum Analyzer is a frequency-domain instrument.

→ Equation of AM wave, AM → Amplitude modulated wave.

$$S(t) = x(t) \cos \omega_c t + A \cos \omega_c t \quad (\text{in time-domain})$$

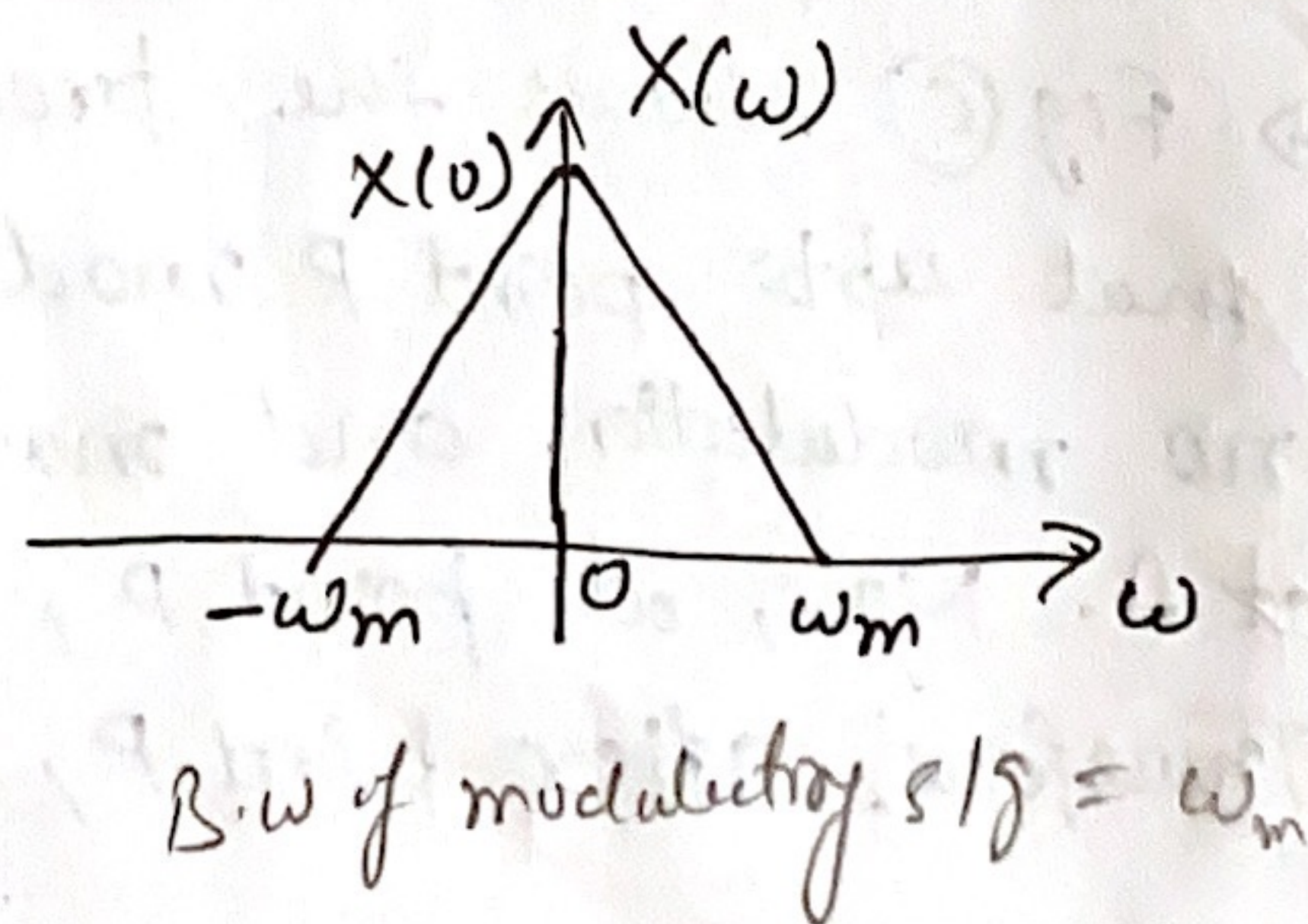
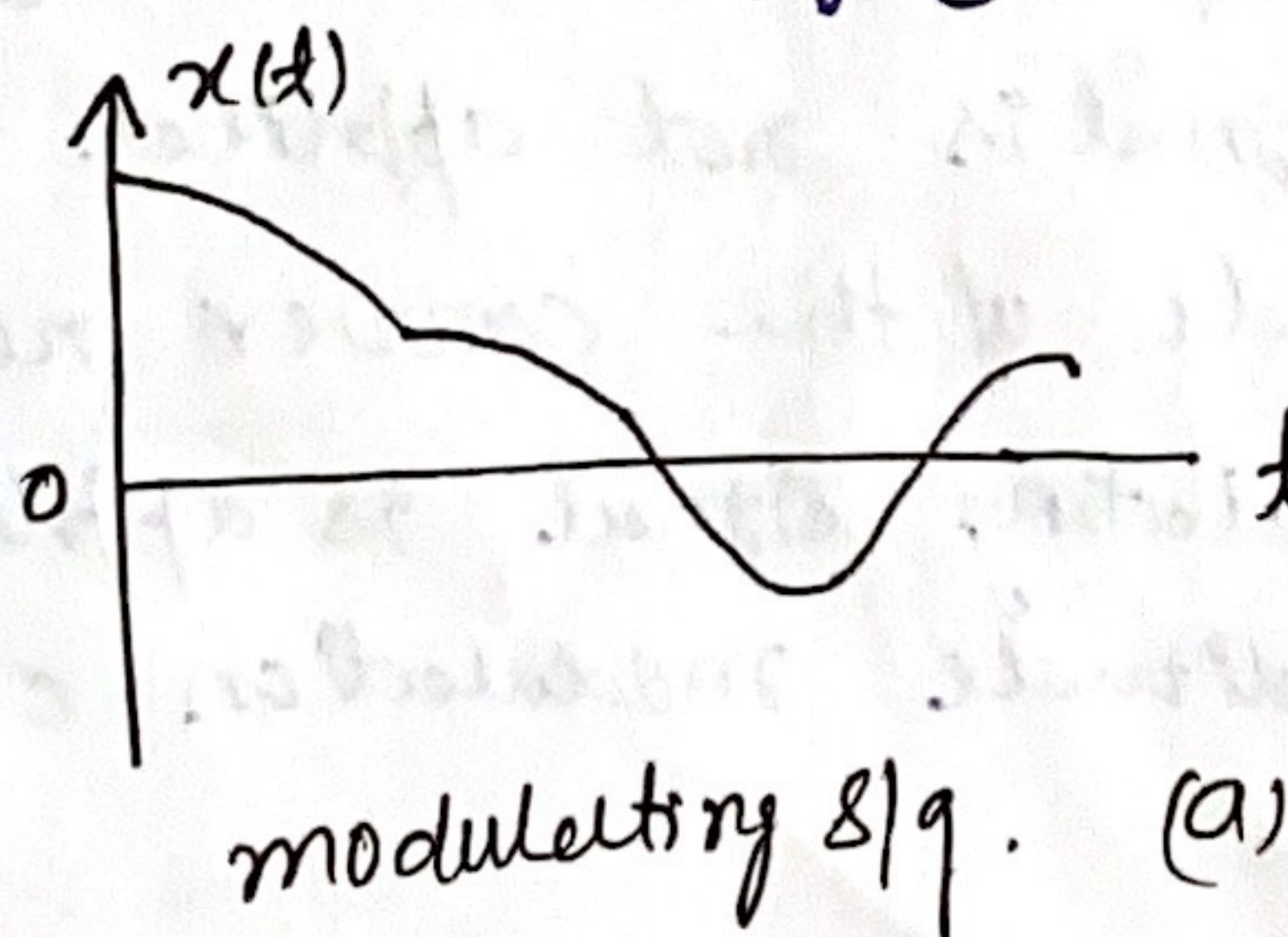
✓ If we want to know the frequencies present in the AM wave, we will have to find its spectrum or frequency-domain representation. For this, we have to take the Fourier transform of AM wave.

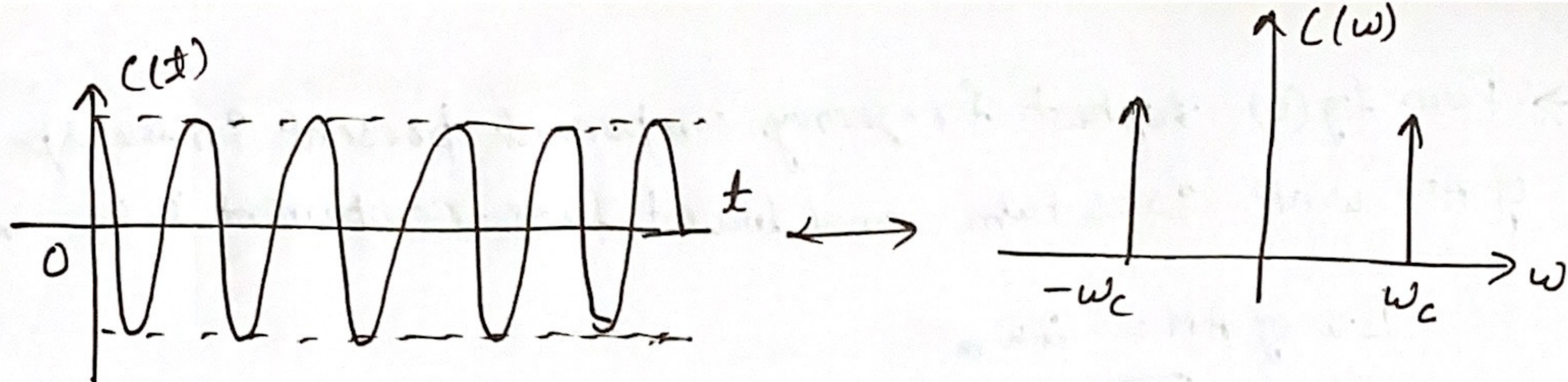
Let $S(\omega) =$ Fourier transform of $S(t)$

$C(\omega) =$ Fourier transform of $C(t)$

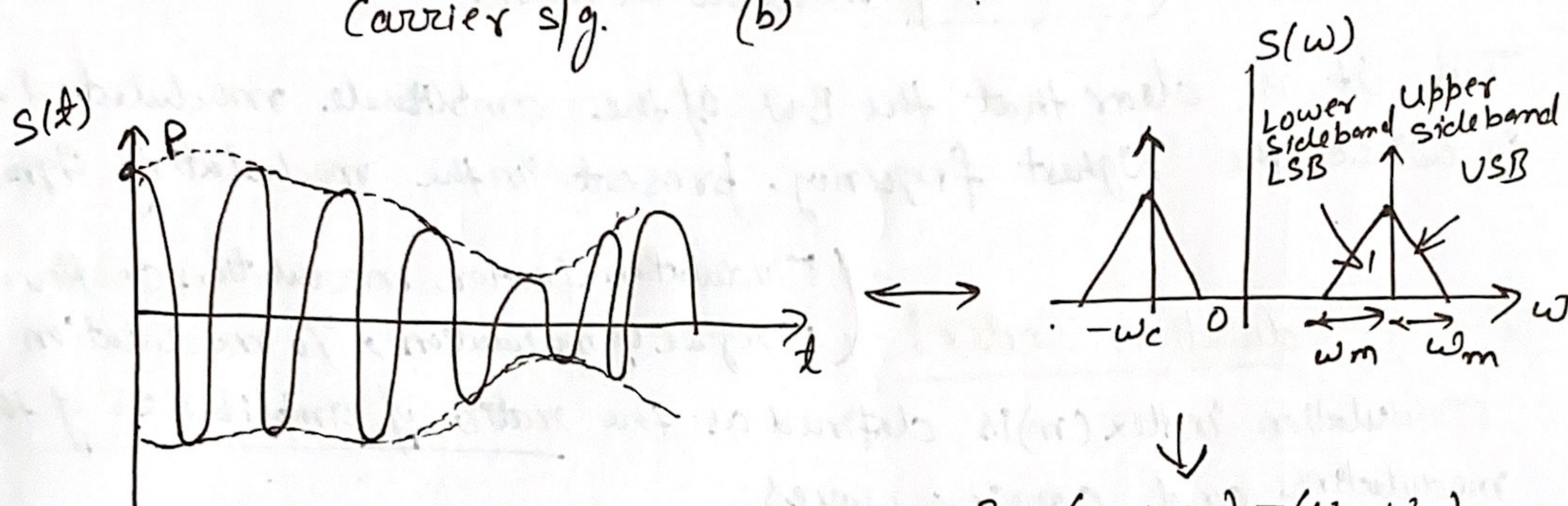
$X(\omega) =$ Fourier transform of $X(t)$

Let the modulating signal $x(t)$ and its Fourier transform $X(\omega)$ be as shown in fig. (a).





Carrier s/g. (b)



modulated wave. (c)

$$B = (\omega_c + \omega_m) - (\omega_c - \omega_m)$$

$$B = 2\omega_m$$

upper and lower freq limit

- As shown in fig (a) modulating signal is band-limited to the interval $-\omega_m \leq \omega \leq \omega_m$, it means modulating signal does not have any freq. component outside the interval $(-\omega_m, \omega_m)$
- In fig (a) modulating signal frequency ranges extend from $-\omega_m$ to ω_m i.e. it also includes -ve frequencies from $-\omega_m$ to 0. Practically there is no meaning of -ve frequency and is used for mathematical convenience only.
- Hence modulating s/g contains frequencies from 0 to ω_m it means B.W of the modulating s/g is ω_m
- Fourier transform of AM wave is given by,

$$S(\omega) = \frac{1}{2} [X(\omega - \omega_c) + X(\omega + \omega_c)] + \pi A [\delta(\omega + \omega_c) + \delta(\omega - \omega_c)]$$

modulatory signal shifted in the +ve as well as in the -ve direction by the factor ω .

carrier signal

→ From fig (c) highest frequency component present in the spectrum of AM wave is $\omega_c + \omega_m$ and lowest freq. component is $\omega_c - \omega_m$

$$\text{B.W of AM} = 2\omega_m$$

$$\boxed{B = 2\omega_m} \text{ modulated wave.}$$

Thus, it is clear that the B.W of the amplitude modulated wave is twice the highest frequency present in the modulating signal.

☆ Modulation Index :- (Modulation factor, modulation coefficient, Degree of modulation, % modulation)

Modulation index (m) is defined as the ratio of amplitudes of the modulating and carrier waves

$$m = \frac{\text{Amplitude of the modulating signal}}{\text{Amplitude of the carrier signal.}}$$

$$\boxed{m = \frac{E_m}{E_c}}$$

$m < 1$ there will be no distortion in AM wave
 $m > 1$ there will be distortion in AM wave

→ When $E_m \leq E_c$, the modulation index 'm' has values b/w 0 and 1 and no distortion is introduced in the AM wave.

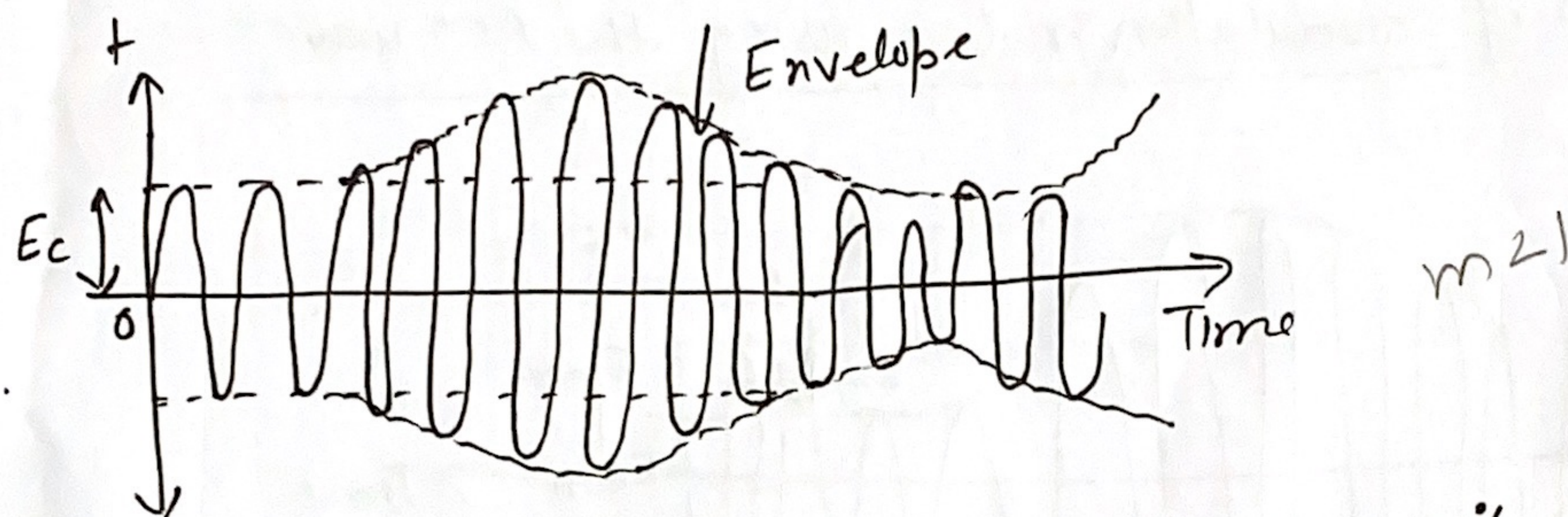
→ When $E_m > E_c$, then m is greater than 1. This will distort the shape of AM signal. The distortion is called as 'overmodulation'.

$$\text{Percentage Modulation} = \frac{E_m}{E_c} \times 100$$

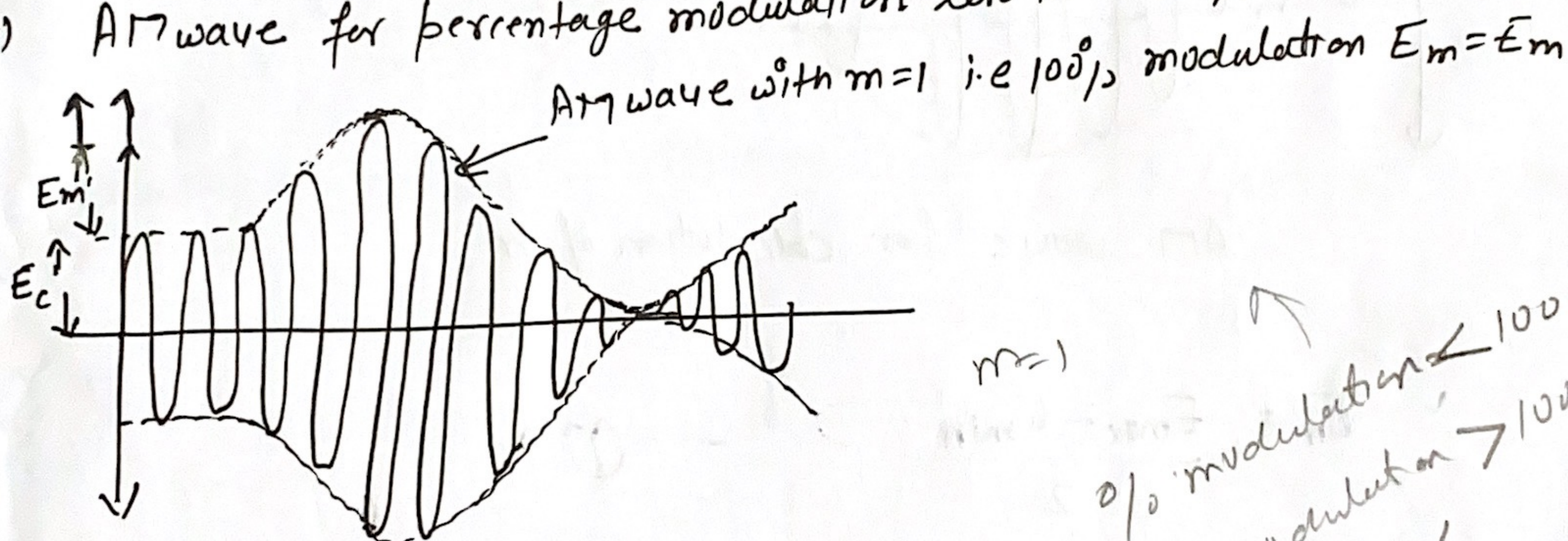
☆ Linear Modulation and Over modulation

(i) Linear Modulation :- (under modulation)

If $m \leq 1$ or if percentage modulation is less than 100, then the type of modulation is linear amplitude modulation.



(a) AM wave for percentage modulation less than 100%.

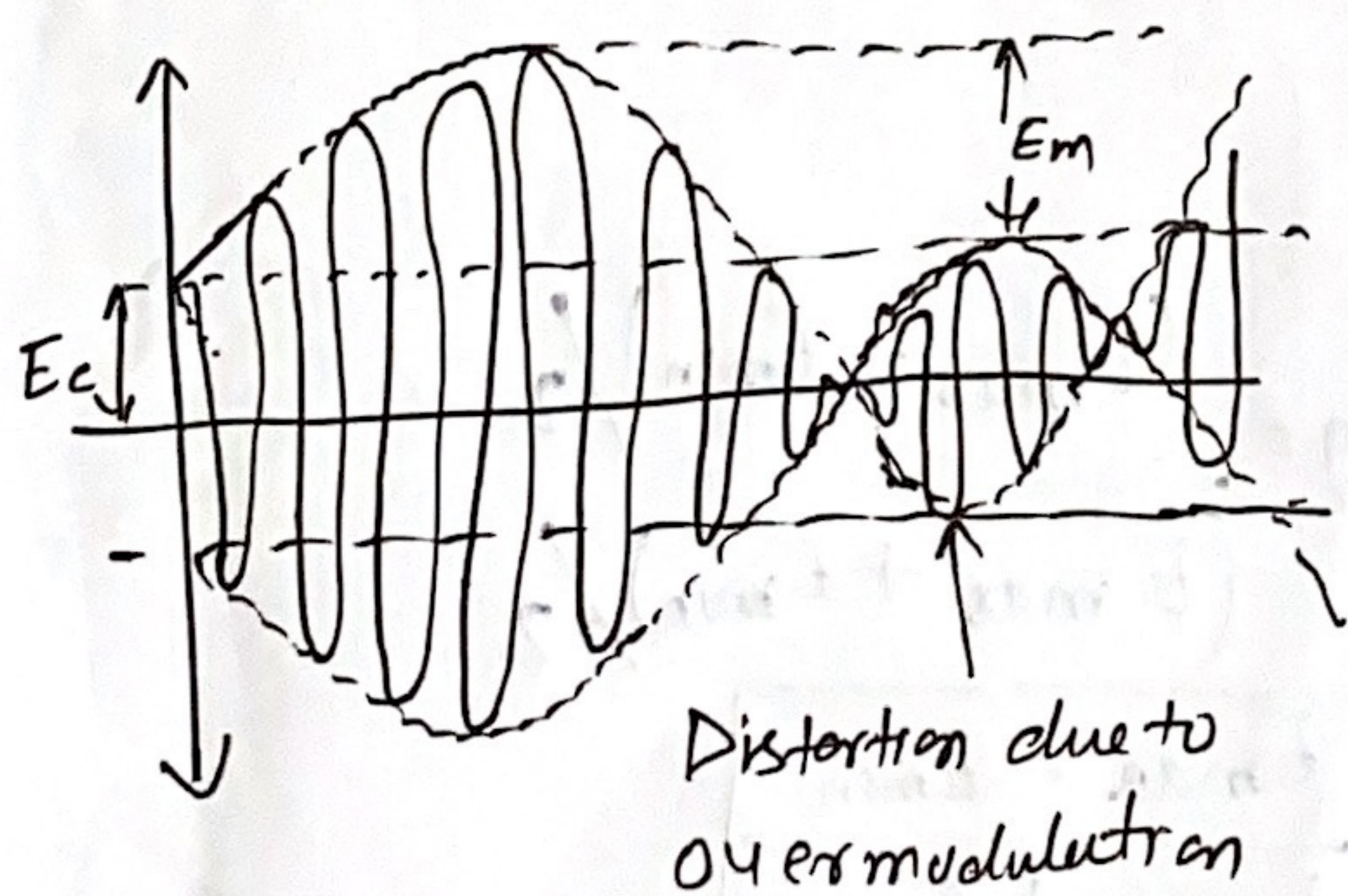


(b) AM wave with 100% modulation

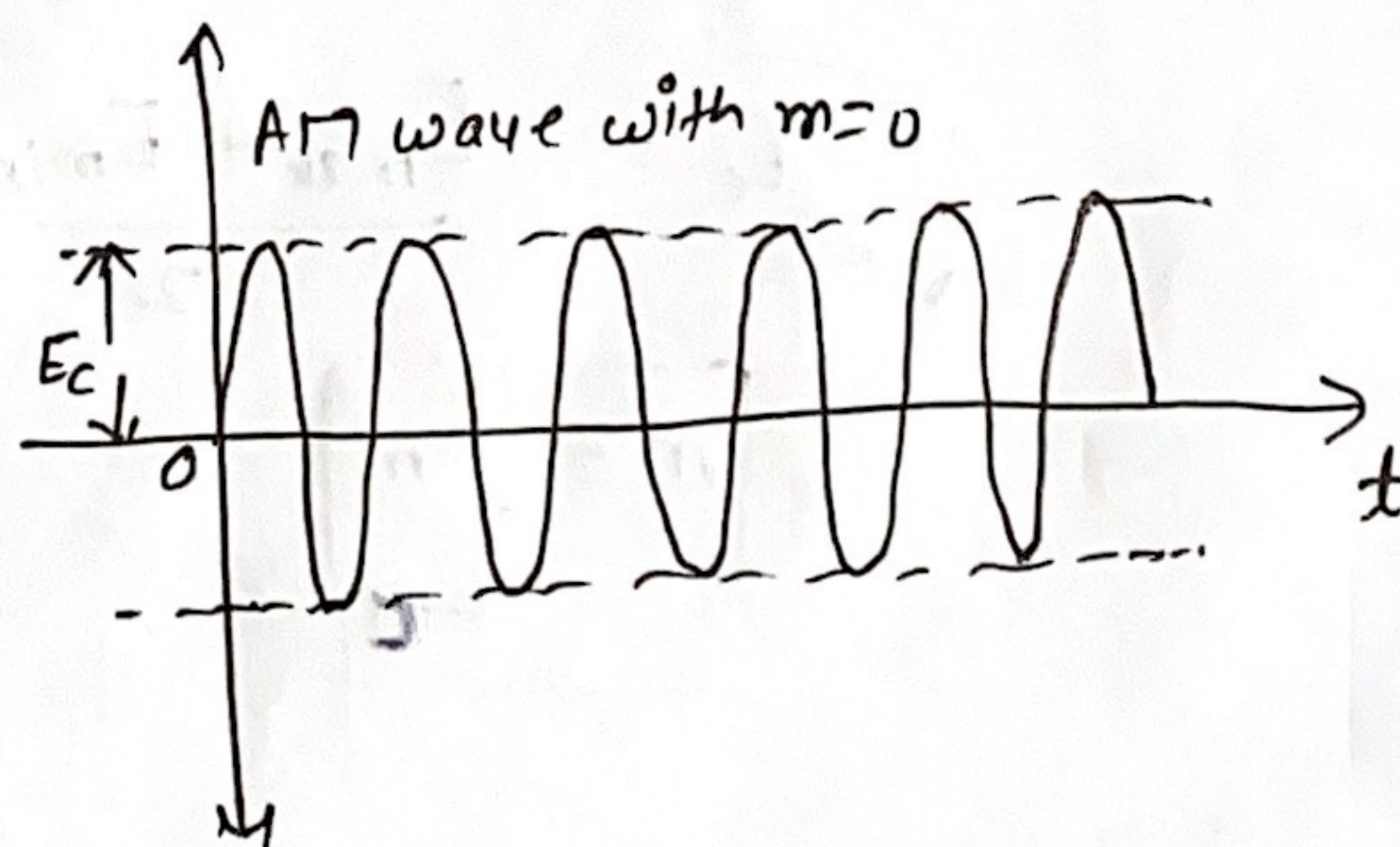
(ii) Over Modulation

If $m > 1$, if the percentage modulation is greater than 100%, then the type of modulation is called as over modulation. Overmodulation introduces envelope distortion. Therefore, it should be avoided.

AM wave with $m > 1$ i.e overmodulation ($E_m > E_c$)



(c) AM wave with over modulation

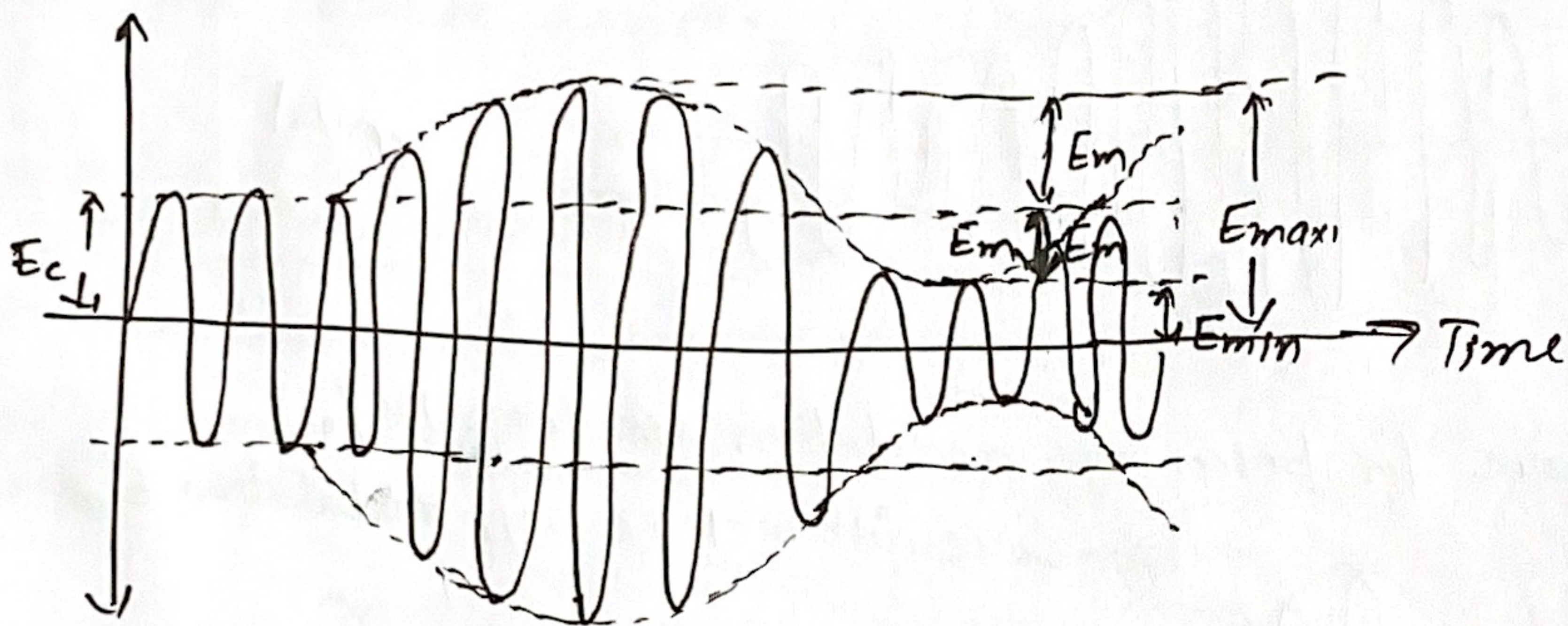


(d) AM wave with $m = 0$

Note Why should the shape of envelope be same as that of modulating signal?

The shape of the AM wave envelope should be same as that of the modulating signal since this reduces the complexity of the demodulator circuit to great extents.

★ Evaluation of modulation index using the AM wave



AM wave for calculation of m

$$E_m = \frac{E_{max} - E_{min}}{2} \quad \text{--- (1)}$$

$$E_c = E_{max} - E_m \quad \text{--- (2)}$$

$$\begin{aligned} E_c &= E_{max} - \left[\frac{E_{max} - E_{min}}{2} \right] \\ &= E_{max} - \frac{E_{max}}{2} + \frac{E_{min}}{2} \end{aligned}$$

$$E_c = \frac{E_{max} + E_{min}}{2}$$

$$m = \frac{E_m}{E_c}$$

$$m = \frac{(E_{max} - E_{min})/2}{(E_{max} + E_{min})/2}$$

$$m = \frac{E_{max} - E_{min}}{E_{max} + E_{min}}$$

★ Transmission Efficiency:-

Transmission efficiency of an AM wave is the ratio of the transmitted power which contains the information (i.e. the total sideband power) to the total transmitted power.

$$\text{Transmission efficiency of AM} = \frac{\text{transmitted power which contains the information}}{\text{total transmitted power}}$$

$$\eta = \frac{P_{LSB} + P_{USB}}{P_t} = \frac{\left[\frac{m^2}{4} P_c + \frac{m^2}{4} P_c \right]}{\left[1 + \frac{m^2}{2} \right] P_c}$$

$$\eta = \frac{\frac{m^2}{2}}{1 + \frac{m^2}{2}} = \frac{m^2}{2 + m^2}$$

Percentage transmission efficiency is given by,

$$\eta = \frac{m^2}{2 + m^2} \times 100\%$$

$$\rightarrow \boxed{P_t = P_c \left(1 + \frac{m^2}{2} \right)}$$

With increased value of m , the transmitted power P_t by the AM transmitter will increase as,

$$P_t = P_c \left(1 + \frac{m^2}{2} \right)$$

☆ Single tone amplitude Modulation (AM):-

- In amplitude modulation we assumed that the modulating signal is a random signal which contains a large number of frequency components. This means that a carrier signal (fixed frequency signal) is modulated by a large number of frequency components.
- But in single tone amplitude modulation, modulating s/g contains only one frequency. i.e. modulation is done by single frequency.

Mathematical Expression

Let us consider a single tone modulating s/g as,

$$x(t) = V_m \cos \omega_m t \quad \text{--- (1)}$$

which contains a single freq. ω_m

carrier signal,

$$c(t) = A \cos \omega_c t \quad \text{--- (2)}$$

general expression for AM signal is,

$$s(t) = [A + x(t)] \cos \omega_c t$$

$$s(t) = A \cos \omega_c t + x(t) \cos \omega_c t \quad \text{--- (3)}$$

Putting the value of eq (1) in eq (3)

$$s(t) = A \cos \omega_c t + V_m \cos \omega_m t \cos \omega_c t$$

$$s(t) = A \cos \omega_c t \left[1 + \frac{V_m}{A} \cos \omega_m t \right] \quad \text{--- (4)}$$

AM modulation index, m_a ,

$$m_a = \frac{|x(t)|_{\max}}{A}$$

$$= \frac{\text{max. amplitude of the modulating s/g}}{\text{max. amplitude of the carrier s/g}}$$

where, $|x(t)|_{\max}$ denotes the maximum amplitude of modulating s/g and A is the max. amplitude of the carrier s/g.

In this case,
(single tone) $|x(t)|_{\max} = V_m$

$$m_a = \frac{V_m}{A}$$

Putting the value of m_a in eq (4) we get

$$S(t) = A \cos \omega_c t [1 + m_a \cos \omega_m t] \quad \text{--- (5)}$$

This is the desired expression for single-tone modulating s/g.

→ The expression in eq (5) may further be simplified to observe the frequency components present in AM signal.

$$S(t) = A \cos \omega_c t [1 + m_a \cos \omega_m t]$$

$$S(t) = A \cos \omega_c t + A \cdot m_a \cos \omega_c t \cos \omega_m t$$

$$S(t) = A \cos \omega_c t + \frac{A \cdot m_a}{2} [2 \cos \omega_c t \cos \omega_m t]$$

$$S(t) = A \cos \omega_c t + \frac{A \cdot m_a}{2} [\cos(\omega_c + \omega_m)t + \cos(\omega_c - \omega_m)t]$$

$$S(t) = A \cos \omega_c t + \frac{A \cdot m_a}{2} \cos(\omega_c + \omega_m)t + \frac{A \cdot m_a}{2} \cos(\omega_c - \omega_m)t \quad \text{--- (6)}$$

eq (6) shows that AM signal has three frequency components.

(i) Carrier frequency ω_c having amplitude A .

(ii) Upper sideband $(\omega_c + \omega_m)$ having amplitude $\frac{m_a \cdot A}{2}$

(iii) Lower sideband $(\omega_c - \omega_m)$ having amplitude $\frac{m_a \cdot A}{2}$

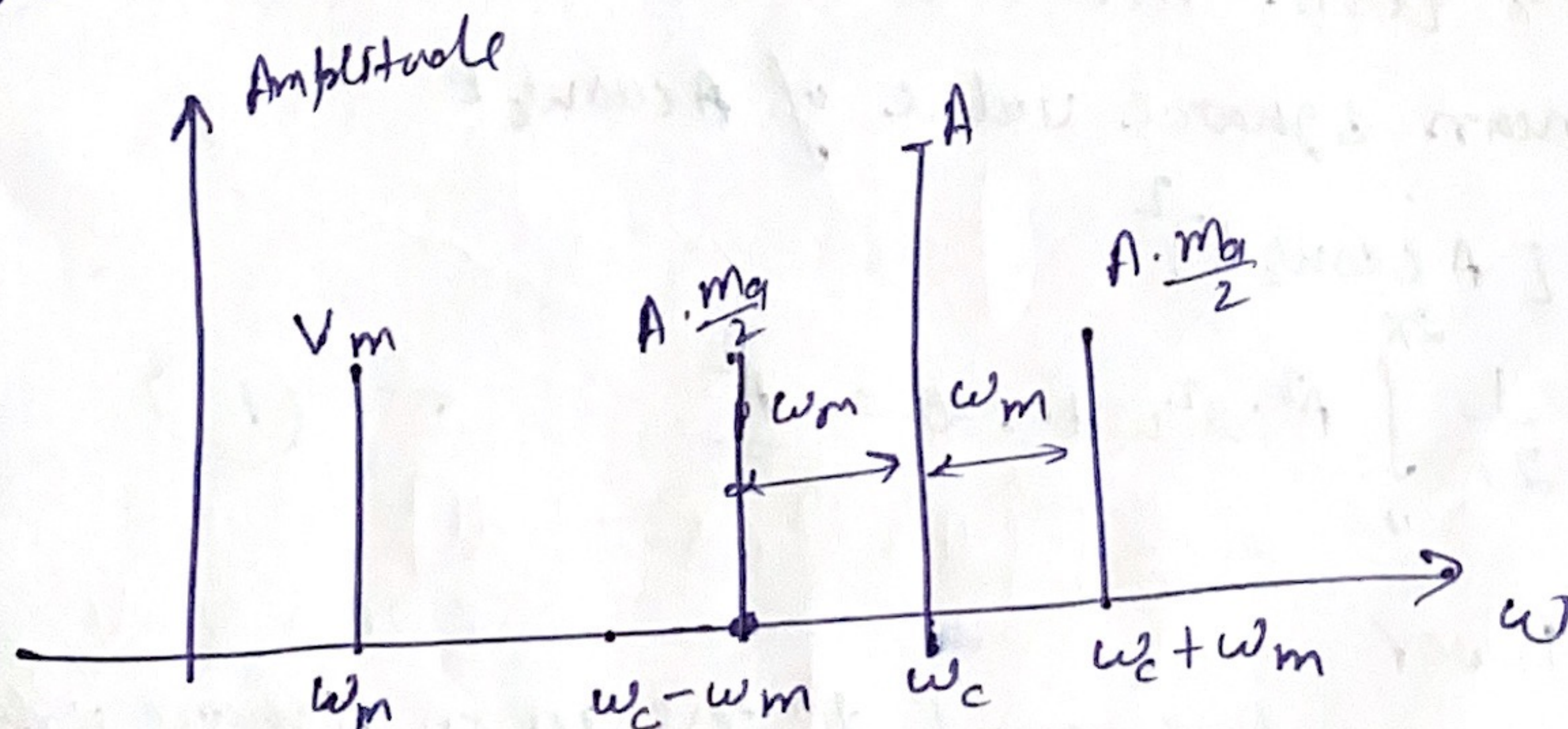


Fig single-sided frequency spectrum of single-tone AM wave.

☆ Power Content in AM wave

→ Modulated wave consists of carrier component and two sideband components.

→ It means that the modulated wave contains more power than the unmodulated carrier.

→ However, since the amplitudes of two sidebands depend upon the modulation index, it may be anticipated that the total power of the amplitude modulated wave would depend upon the modulation index also.

⇒ To find the power contents of the carrier and the sidebands,

General expression of the AM wave,

$$S(t) = A \cos \omega_c t + x(t) \cos \omega_c t$$

✓ The total power P of the AM wave is the sum of the carrier power P_c and sideband power P_s ,

* Since period of the signal $A \cos \omega_c t$ is 2π .

① Carrier Power

The carrier power P_c is equal to the mean-square value of the carrier term $A \cos \omega_c t$ i.e.

$P_c = \text{mean square value of } A \cos \omega_c t$

$$P_c = [A \cos \omega_c t]^2$$

$$= \frac{1}{2\pi} \int_0^{2\pi} A^2 \cos^2 \omega_c t \cdot dt = \frac{A^2}{2}$$

$$= \frac{A^2}{4\pi} \left[\int_0^{2\pi} dt + \int_0^{2\pi} \cos 2\omega_c t \cdot dt \right]$$

$$= \frac{A^2}{4\pi} [2\pi + \sin 4\pi \omega_c t]_0^{2\pi}$$

② Sideband Power

The sideband power P_s is equal to the mean square value of the sideband term $x(t) \cos \omega_c t$ i.e.

$P_s = \text{mean square value of } x(t) \cos \omega_c t$

$$P_s = [x(t) \cos \omega_c t]^2 = \frac{1}{2\pi} \int_0^{2\pi} x^2(t) \cos^2 \omega_c t \cdot dt = \frac{1}{2\pi} \int_0^{2\pi} \frac{1}{2} [2 \cos^2 \omega_c t] x^2(t) dt$$

$$\text{or } P_s = \frac{1}{2\pi} \int_0^{2\pi} \frac{1}{2} x^2(t) dt + \frac{1}{2} \int_0^{2\pi} x^2(t) \cos 2\omega_c t dt$$

- In AM generation, a Band pass Filter (BPF) or a tuned circuit tuned to carrier freq ω_c is used to filter out the second integral term.

$$P_s = \frac{1}{2\pi} \int_0^{2\pi} \left[\frac{1}{2} x^2(t) \right] dt$$

$$\text{or } P_s = \text{mean square value of } \frac{1}{2} x^2(t) = \frac{1}{2} \overline{x^2(t)}$$

→ the total sideband power P_s is due to the equal contributions of the upper and lower sidebands, hence, the power carried by the upper and the lower sideband will be.

$$P_{s(\text{LSB})} = P_{s(\text{USB})} = \frac{P_s}{2} = \frac{1}{4} \overline{x^2(t)}$$

→ Therefore, the total power P_t of the AM signal is the sum of the carrier power P_c and sideband power P_s .

$$\begin{aligned} P_t = P_c + P_s &= \frac{1}{2} A^2 + \frac{1}{2} \overline{x^2(t)} \\ &= \frac{1}{2} [A^2 + \overline{x^2(t)}] \end{aligned}$$

$$\boxed{P_t = \frac{1}{2} [A^2 + \overline{x^2(t)}]}$$